

PakTradeX

AI-Powered Pakistan Capital Market Platform

PROJECT FEATURE CHECKLIST & AUDIT REPORT

Generated: 24 August 2026, 03:25 PM • Repo: github.com/14879-dev/PakTradeX • Author: 14879-dev

153

Completed

11

In Progress

70

Pending

65%

Complete

1. PROJECT SETUP & INFRASTRUCTURE

Repository, CI/CD, DevOps, Environment

■ Flutter project created (paktradex, v1.0.0+1)	DONE
■ Git repository initialized → GitHub: github.com/14879-dev/PakTradeX	DONE
■ Git identity configured: 14879-dev / 14879@cityuniversity.edu.pk	DONE
■ Android package name set: com.paktradex.app	DONE
■ FastAPI backend scaffolded: /backend/ directory	DONE
■ SQLite database (dev): paktradex.db auto-initialized	DONE
■ Dockerfile created for containerized deployment	DONE
■ Procfile created for Heroku/Railway	DONE
■ railway.json configured for Railway deploy	DONE
■ render.yaml configured for Render deploy	DONE
■ docker-compose.yml for local multi-service dev	DONE
■ requirements.txt with all Python dependencies pinned	DONE
■ .env + .env.example with all config keys documented	DONE
■ backend/.gitignore excludes .env and paktradex.db	DONE
■ localtunnel HTTPS tunnel: paktradex-mm521142.loca.lt	DONE
■ Windows Firewall script: scripts/open_firewall_port_8000.ps1	DONE
■■ Railway/Render cloud deployment — NOT completed (no card)	PARTIAL
■ GitHub Actions CI/CD pipeline not set up	TODO
■ PostgreSQL production database not configured	TODO
■ Domain name / custom SSL not configured	TODO

2. FLUTTER DEPENDENCIES & THEME

Packages, Design System, Typography, Colors

■ flutter_riverpod ^2.6.1 — state management	DONE
■ go_router ^14.8.1 — declarative routing	DONE
■ dio ^5.8.0+1 — HTTP client with interceptors	DONE
■ flutter_secure_storage ^9.2.4 — JWT token storage	DONE
■ google_fonts ^6.2.1 — Inter/Outfit typography	DONE
■ fl_chart ^0.70.2 — financial charts	DONE
■ intl ^0.20.2 — number/date formatting	DONE
■ AppColors — full dark-mode-aware color palette	DONE
■ AppTypography — financial display + body text scales	DONE
■ AppSpacing — 4px grid spacing system	DONE
■ AppRadius — consistent border radius tokens	DONE
■ AppCard, PrimaryButton, SecondaryButton, SectionHeader widgets	DONE
■ PriceChangeBadge widget with color-coded $\pm\%$ display	DONE
■ local_auth package not added (biometric login stub only)	TODO
■ flutter_local_notifications not added	TODO
■ Dark mode theme switcher not implemented	TODO

3. BACKEND API — AUTHENTICATION

FastAPI endpoints: /api/v1/auth/*

■	POST /auth/register — creates user, sends OTP email	DONE
■	POST /auth/login — validates credentials, sends 2FA OTP	DONE
■	POST /auth/verify-otp — validates OTP, issues 7-day JWT	DONE
■	POST /auth/resend-otp — generates & resends OTP	DONE
■	POST /auth/forgot-password — sends reset OTP	DONE
■	GET /auth/me — returns user profile from JWT	DONE
■	bcrypt password hashing via passlib	DONE
■	HS256 JWT issuance (7-day expiry) via python-jose	DONE
■	In-memory OTP store (_pending_otp dict)	DONE
■	Duplicate email 409 Conflict response	DONE
■	PTX-XXXXXX unique ID generation per user	DONE
■	SMTP email service (Gmail smtp.gmail.com:587)	DONE
■	OTP fallback logging when SMTP not configured	DONE
■■	SMTP_PASSWORD = PASTE_APP_PASSWORD_HERE (not yet filled)	PARTIAL
■	OTP expiry/TTL (currently valid indefinitely in memory)	TODO
■	Rate limiting on OTP endpoints	TODO
■	Password reset confirm + change endpoint	TODO
■	Refresh token endpoint	TODO
■	Social auth (Google/Facebook OAuth)	TODO

4. BACKEND API — MARKETS & TRADING

FastAPI endpoints: /api/v1/markets/* and /api/v1/trading/*

■	GET /markets/quotes — stock quotes with yfinance fallback	DONE
■	GET /markets/quote/{symbol} — individual stock quote	DONE
■	GET /markets/overview — KSE-100/KSE-30 index summary	DONE
■	GET /markets/gainers — top gainer stocks	DONE
■	GET /markets/losers — top loser stocks	DONE
■	GET /markets/order-book/{symbol} — synthetic order book	DONE
■	POST /trading/order — place buy/sell order	DONE
■	GET /trading/portfolio — holdings + available cash	DONE
■	GET /trading/orders — trade history	DONE
■	PSX brokerage fee: 0.15% (min Rs.25, max Rs.500)	DONE
■	Demo portfolio: Rs.1,000,000 virtual capital	DONE
■	SQLite persistence for portfolios/holdings/orders	DONE
■ ■	yfinance data uses .NS suffix fallback (PSX not on Yahoo)	PARTIAL
■	Real PSX data feed (PMEX / NCCPL API) not integrated	TODO
■	Limit orders not implemented (market orders only)	TODO
■	Stop-loss / take-profit orders not implemented	TODO
■	Real money / live trading not implemented	TODO
■	P2P Transfer backend endpoint not implemented	TODO
■	Deposit/Withdrawal backend not implemented	TODO

5. BACKEND API — AI COPILOT & NEWS

FastAPI endpoints: /api/v1/copilot/* and /api/v1/news/*

■	POST /copilot/chat — Gemini AI financial analysis	DONE
■	Gemini 1.5 Flash integration via google-generativeai	DONE
■	PSX-context system prompt for Urdu/English bilingual response	DONE
■	GET /news/ — news articles endpoint	DONE
■	Sentiment analysis (bullish/bearish/neutral) in AI response	DONE
■ ■	News endpoint returns mock data (no RSS/scrapper integrated)	PARTIAL
■	Real news API (Bloomberg/Dawn/BusinessRecorder) not integrated	TODO
■	Stock-specific news filtering not implemented	TODO
■	AI portfolio analysis endpoint not implemented	TODO

6. FLUTTER — AUTHENTICATION SCREENS

lib/features/auth/presentation/

■ SplashScreen — animated logo + auto-redirect with session restore	DONE
■ OnboardingScreen — 3-slide animated onboarding carousel	DONE
■ OnboardingScreen — Skip button routes to /login	DONE
■ LoginScreen — email + password fields (no demo pre-fill)	DONE
■ LoginScreen — forgot password link, register redirect	DONE
■ RegisterScreen — full name, email, phone, password fields	DONE
■ OtpVerificationScreen — 6-digit PIN input with countdown	DONE
■ OtpVerificationScreen — resend OTP button	DONE
■ ForgotPasswordScreen — email input + backend OTP trigger	DONE
■ Auth Riverpod provider with full state machine	DONE
■ JWT token saved to flutter_secure_storage	DONE
■ Session restore on app launch (tryBiometricUnlock)	DONE
■ Critical bug fixed: email captured before state.authenticating()	DONE
■ Biometric (fingerprint/face) actual unlock not implemented	TODO
■ Password visibility toggle on login not working on some devices	TODO
■ Google Sign-In not implemented	TODO
■ Password reset completion (enter new password) not implemented	TODO

7. FLUTTER — HOME SCREEN

lib/features/home/presentation/

■ HomeScreen scaffold with greeting + balance header	DONE
■ PortfolioSummaryCard — total value, daily P&L;	DONE
■ MarketOverviewCard — KSE-100, KSE-30, KMI-30 index tiles	DONE
■ MarketMoversCard — Top Gainers / Top Losers tabbed list	DONE
■ RecentNewsCard — latest 3 news articles	DONE
■ AiMarketBriefCard — daily AI market summary	DONE
■ Pull-to-refresh on home screen	DONE
■ Mock market data (30 PSX stocks) for offline use	DONE
■ ■ Backend market data connected but uses mock fallback	PARTIAL
■ Real-time price polling (30s) not wired to live backend data	TODO
■ Push notifications for price alerts not implemented	TODO

8. FLUTTER — MARKETS SCREEN		lib/features/markets/presentation/
■	MarketsScreen — 6 tabs: Insights, Favorites, PSX Stocks, Gainers, Shariah, Sectors	DONE
■	Insights tab — Fear & Greed gauge, 24H volume, Bulls/Bears ratio	DONE
■	Insights tab — Featured stock card (OGDC) with sparkline	DONE
■	Insights tab — Category mini cards (Banks/Tech/Energy)	DONE
■	Insights tab — AI For You banner → /ai route	DONE
■	Insights tab — Rs.1,000,000 Trading Gala promotional banner	DONE
■	PSX Stocks tab — searchable, sector-filterable list	DONE
■	Favorites tab — starred stocks list	DONE
■	Shariah (KMI) tab — filtered halal stock list	DONE
■	Sector filter chips (horizontal scrollable)	DONE
■	Search bar with live filtering	DONE
■ ■	RenderFlex overflow by 81px on right (pending fix)	PARTIAL
■	Favorites stored in local persistence (not in memory)	TODO
■	Real sector data from PSX API not integrated	TODO
■	QR scanner for PakTrade ID payments not implemented	TODO

9. FLUTTER — STOCK DETAIL SCREEN		lib/features/stock_details/presentation/
■	StockDetailScreen — header with logo, price, change badge	DONE
■	Interactive sparkline chart with time range tabs (1D/1W/1M/3M/1Y)	DONE
■	Order book widget (bids/asks)	DONE
■	Key metrics: P/E, Dividend Yield, Market Cap, Volume	DONE
■	Buy/Sell action buttons → TradeScreen	DONE
■	About section with company description	DONE
■	Real OHLCV chart data not wired to backend	TODO
■	Technical indicators (RSI, MACD, Bollinger) not implemented	TODO
■	News related to symbol not shown	TODO
■	Watchlist add/remove from detail screen not implemented	TODO

10. FLUTTER — TRADING SCREEN		lib/features/trading/presentation/
■	TradeScreen — buy/sell toggle, quantity input, price preview	DONE
■	OrderSheet bottom sheet modal — order summary + confirm	DONE
■	Brokerage fee calculation: 0.15% (Rs.25 min, Rs.500 max)	DONE
■	Demo mode: Rs.1,000,000 virtual capital label	DONE
■	Trade order sent to backend POST /trading/order	DONE
■	Account mode switcher (Demo / Real) in profile	DONE
■	Limit order type not implemented (market only)	TODO
■	Order confirmation animation not implemented	TODO
■	Real money trading flow not implemented	TODO

11. FLUTTER — PORTFOLIO SCREEN		lib/features/portfolio/presentation/
■	PortfolioScreen — total value, available cash, invested value	DONE
■	HoldingsList — position cards with unrealized P&L;	DONE
■	OrderHistoryList — executed trade orders	DONE
■	SectorAllocationPieChart — sector breakdown	DONE
■	DepositCashModal — demo balance top-up	DONE
■	P2PTransferModal — PakTrade ID peer transfer UI	DONE
■ ■	P2P Transfer backend not implemented (UI only)	PARTIAL
■	Real portfolio value from backend not connected	TODO
■	Performance graph over time not implemented	TODO
■	Export portfolio as CSV/PDF not implemented	TODO

12. FLUTTER — PROFILE & KYC		lib/features/profile/presentation/
■	ProfileScreen — user info, PakTrade ID, KYC status	DONE
■	Account mode switcher (Demo/Real) with balance display	DONE
■	KYC Verification modal — CNIC/document upload UI	DONE
■	Logout button clears JWT and navigates to /login	DONE
■ ■	KYC upload stub — no backend processing	PARTIAL
■	CNIC verification API (NADRA) not integrated	TODO
■	Profile photo upload not implemented	TODO
■	Change password flow not implemented	TODO
■	2FA enable/disable toggle not implemented	TODO
■	Account deletion not implemented	TODO

13. FLUTTER — AI COPILOT SCREEN lib/features/ai/presentation/	
■ AiCopilotScreen — chat interface with message bubbles	DONE
■ AI message bubble with confidence score + sentiment badge	DONE
■ Action prompt chips below AI response	DONE
■ Related stock symbol tag on AI responses	DONE
■ Bilingual support (English/Urdu prompt toggle)	DONE
■ Connected to backend POST /copilot/chat → Gemini AI	DONE
■ Chat history persistence not implemented	TODO
■ Voice input not implemented	TODO
■ AI portfolio analysis (analyze my portfolio) not implemented	TODO

14. FLUTTER — NEWS SCREEN lib/features/news/presentation/	
■ NewsScreen — article cards with image, headline, source, time	DONE
■ NewsDetailScreen — full article view with share button	DONE
■ Category tabs (All, PSX, Economy, Company News)	DONE
■ Mock news data with realistic PSX headlines	DONE
■ Real news API not integrated (Dawn, BusinessRecorder, Reuters)	TODO
■ News search not implemented	TODO
■ Push notification for breaking news not implemented	TODO

15. FLUTTER — WATCHLIST lib/features/watchlist/	
■■ Watchlist directory exists but screen not fully implemented	PARTIAL
■ Watchlist screen with add/remove stocks not implemented	TODO
■ Price alerts on watchlist items not implemented	TODO

16. APP ROUTING & NAVIGATION

lib/app/router.dart — GoRouter

■ /splash — SplashScreen (initial route)	DONE
■ /onboarding — OnboardingScreen	DONE
■ /login — LoginScreen	DONE
■ /register — RegisterScreen	DONE
■ /otp?target=email — OtpVerificationScreen	DONE
■ /forgot-password — ForgotPasswordScreen	DONE
■ /profile — ProfileScreen	DONE
■ /ai — AiCopilotScreen	DONE
■ /news — NewsScreen + /news/detail sub-route	DONE
■ /home — HomeScreen (shell branch 0)	DONE
■ /home/stock/:symbol — StockDetailScreen	DONE
■ /markets — MarketsScreen (shell branch 1)	DONE
■ /markets/stock/:symbol — StockDetailScreen	DONE
■ /trade — TradeScreen (shell branch 2)	DONE
■ /portfolio — PortfolioScreen (shell branch 3)	DONE
■ StatefulShellRoute bottom nav with persistent state	DONE
■ Deep link handling for notification taps not set up	TODO
■ Route guards for authenticated routes not fully implemented	TODO

17. CORE SERVICES & API CLIENT

lib/core/services/

■ ApiClient — Dio with base URL, timeout, auth interceptor	DONE
■ bypass-tunnel-reminder header for localtunnel	DONE
■ ApiException wrapper with user-friendly error messages	DONE
■ AuthService — register, login, verifyOtp, resendOtp, saveToken	DONE
■ AppConfig — centralized config (API URL, fees, intervals)	DONE
■ API base URL: https://paktradex-mm521142.loca.lt/api/v1	DONE
■ useMockFallback = true (falls back to mock data if backend down)	PARTIAL
■ Offline-first data caching (SharedPreferences/Hive) not implemented	TODO

18. BUGS FIXED DURING SESSION		All confirmed bug fixes committed to main branch
■	Login screen: removed pre-filled demo credentials	DONE
■	Onboarding Skip button: now routes to /login (was /home)	DONE
■	OTP 422 error: email captured before AuthState.authenticating() wipe	DONE
■	Backend import paths: fixed 3-dot → 4-dot relative imports	DONE
■	Removed offline OTP bypass that authenticated without backend	DONE
■	Tunnel URL stabilized: paktradex-mm521142 unique subdomain	DONE
■	Onboarding slide 3: removed 'Simulated Demo' wording	DONE
■	Windows Firewall rule script for port 8000	DONE
■■	RenderFlex overflow 81px on Markets screen — identified, NOT YET FIXED	PARTIAL

19. WHAT TO DO NEXT (PRIORITY ORDER)		Recommended next steps for completion
■	[URGENT] Fill SMTP_PASSWORD in backend/.env → real OTPs to email	TODO
■	[URGENT] Fix RenderFlex overflow on Markets screen	TODO
■	Connect portfolio screen to live backend data	TODO
■	Implement OTP expiry (TTL = 5 minutes)	TODO
■	Add rate limiting on auth endpoints	TODO
■	Implement password reset completion flow	TODO
■	Add watchlist screen with add/remove functionality	TODO
■	Integrate real PSX news API (Dawn Business/Reuters)	TODO
■	Wire real-time quote polling (30s) to live backend	TODO
■	Add local_auth for biometric login	TODO
■	Set up GitHub Actions CI for auto-testing	TODO
■	Deploy to Railway/Render with PostgreSQL (needs card)	TODO
■	Implement NADRA CNIC verification for KYC	TODO
■	Add flutter_local_notifications for price alerts	TODO
■	Dark mode theme switcher	TODO